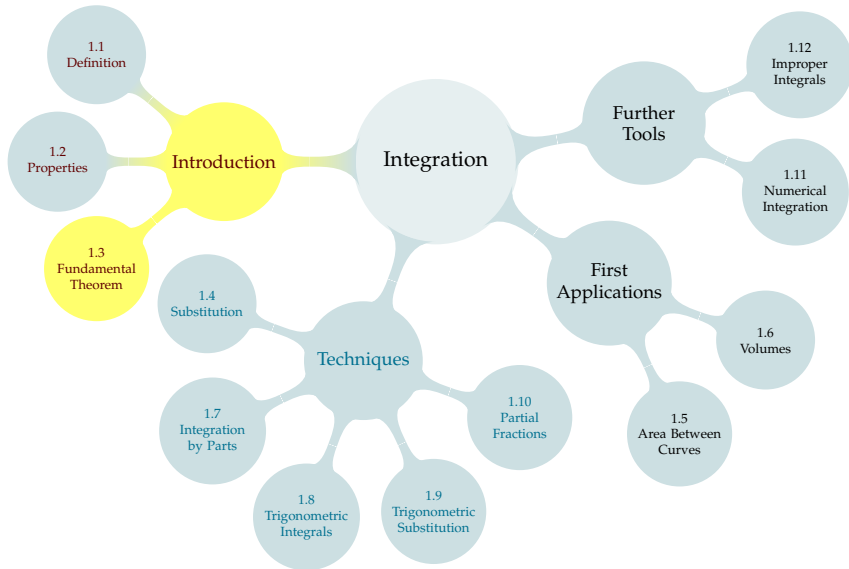


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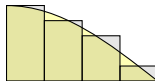
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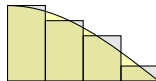
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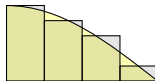
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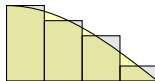
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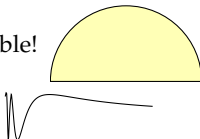
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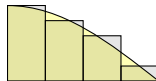
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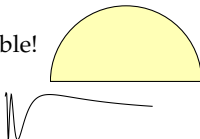
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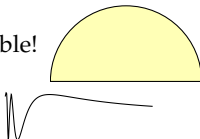
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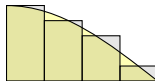
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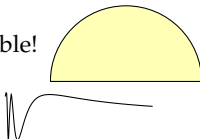
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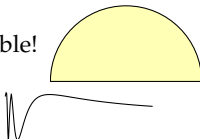
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## Fundamental Theorem of Calculus, Part 1

Let  $a < b$  and let  $f(x)$  be a function which is defined and continuous on  $[a, b]$ . Let

$$A(x) = \int_a^x f(t) \, dt$$

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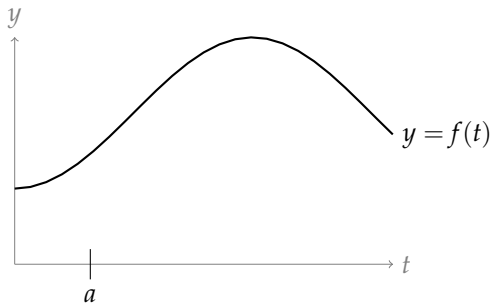
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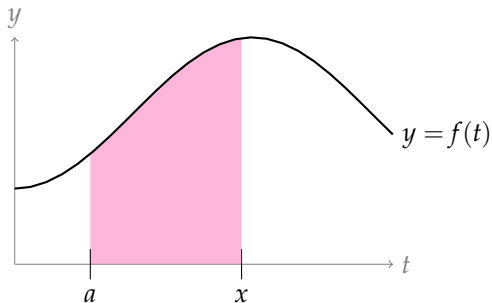
FTC(I) gives us the derivative of a very specific function (subject to some fine print).

It shows a close relationship between integrals and derivatives.

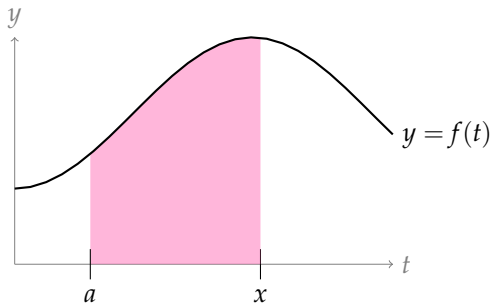
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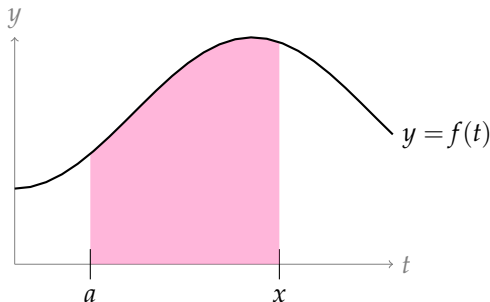
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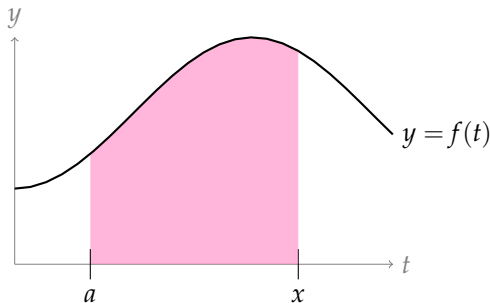
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Notation: the function  $A$  depends on the variable  $x$ .

We need to know how the function  $f$  behaves on the whole interval  $(0, x)$  to find  $A(x)$ . That's why we use  $f(t)$ , not  $f(x)$ .

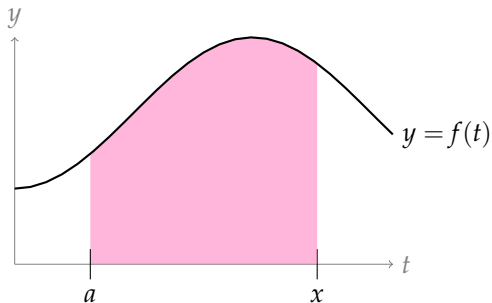
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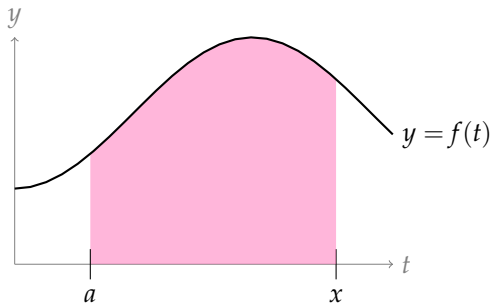
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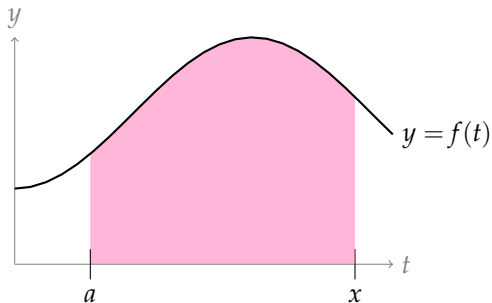
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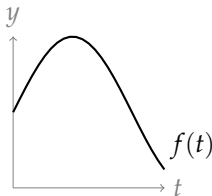
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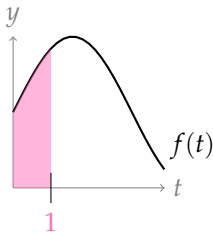


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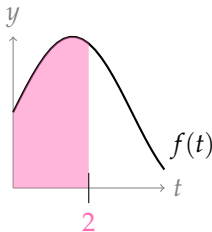
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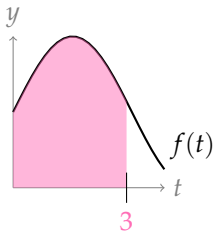
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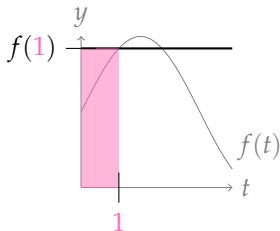
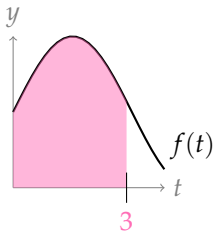
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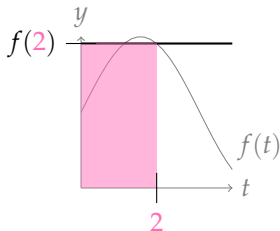
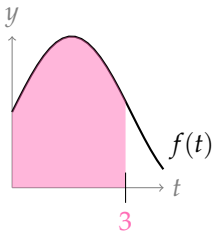
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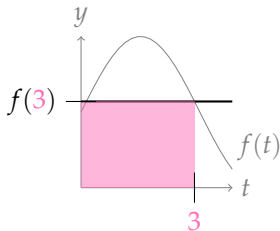
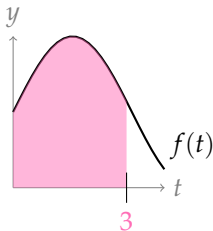
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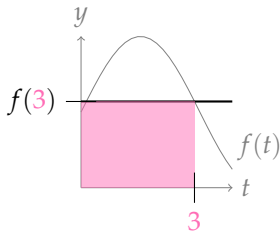
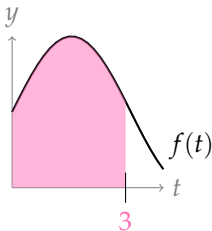
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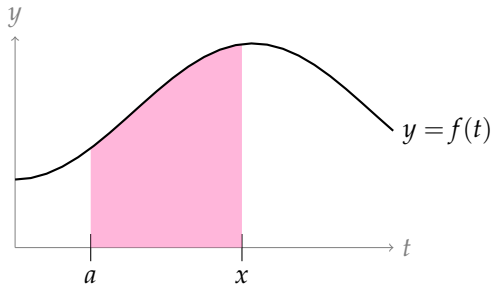
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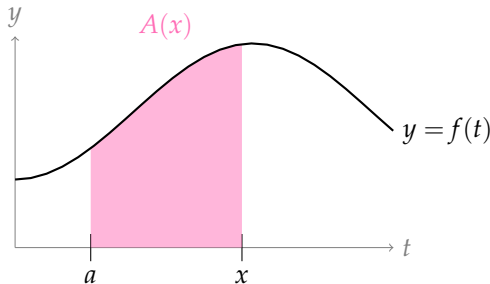
Question: Why is it true?

# DERIVATIVE OF AREA FUNCTION, $A(x) = \int_a^x f(t) dt$



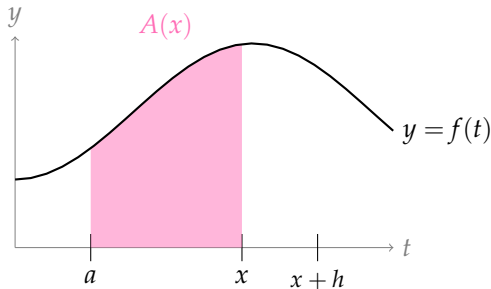
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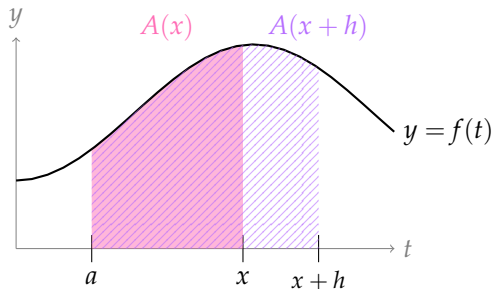
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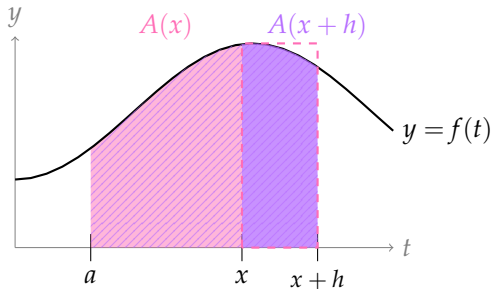
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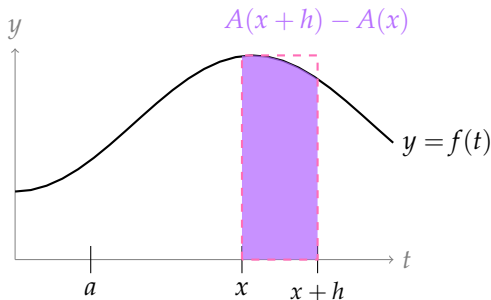
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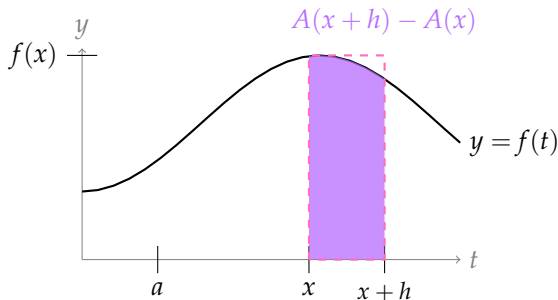
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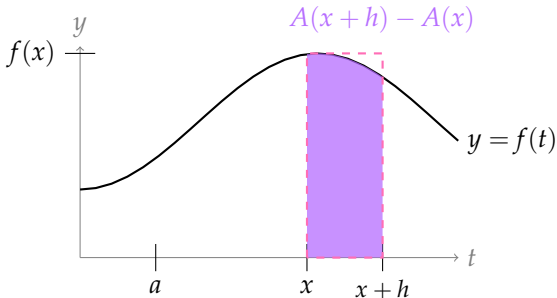
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When  $h$  is very small, the purple area looks like a rectangle with base  $h$  and height  $f(x)$ , so  $A(x+h) - A(x) \approx hf(x)$  and  $\frac{A(x+h) - A(x)}{h} \approx f(x)$ . As  $h$  tends to zero, the error in this approximation approaches 0.

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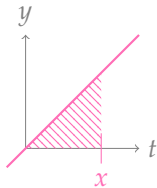
$$A'(x) = f(x) .$$

Suppose  $C(x) = \int_2^{e^x} \sin t \, dt$ . What is  $C'(x)$ ?

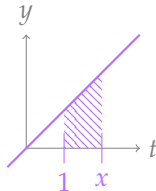


It's possible to have two different functions with the same derivative.

$$A(x) = \int_0^x 2t \, dt$$

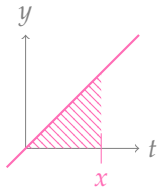


$$B(x) = \int_1^x 2t \, dt$$



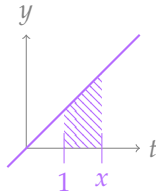
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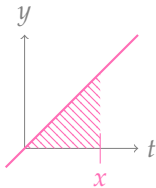
$$A'(x) = 2x$$

$$B(x) = \int_1^x 2t \, dt$$



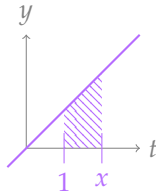
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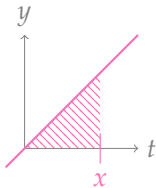
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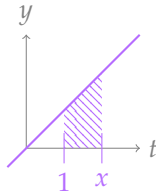
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$$A(x) = \int_0^x 2t \, dt = x^2$$



$$A'(x) = 2x$$

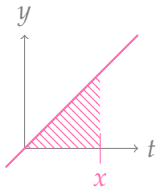
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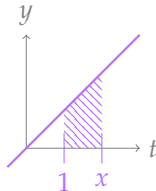
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$$A'(x) = 2x$$

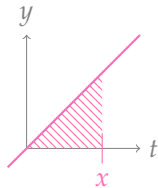
$$B(x) = \int_1^x 2t \, dt = x^2 - 1$$



$$B'(x) = 2x$$

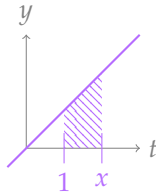
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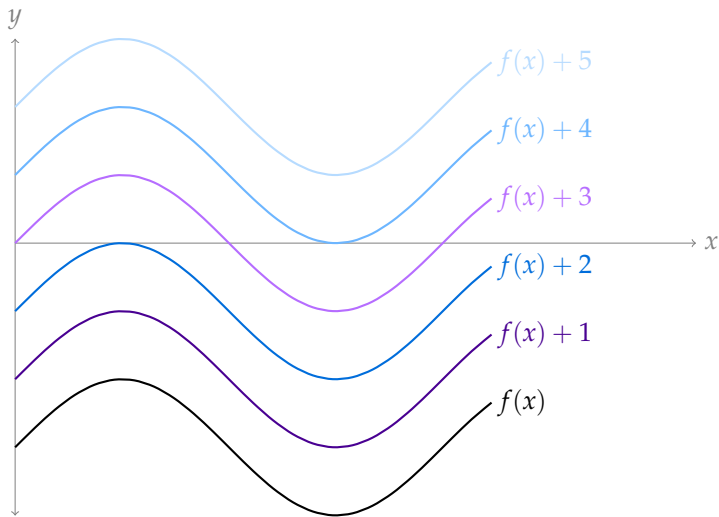
$$B(x) = \int_1^x 2t \, dt = x^2 - 1$$



$$B'(x) = 2x$$

When two functions have the same derivative, they differ only by a constant.

In this example:  $B(x) = A(x) - 1$



If two continuous functions have the same derivative, then one is a constant plus the other.

Two clues for finding  $A(x) = \int_a^x f(t) \, dt$ :

- ▶ If  $A(x) = \int_a^x f(t) \, dt$ , then<sup>1</sup>  $A'(x) = f(x)$
- ▶ If  $F'(x) = A'(x)$ , then  $A(x) = F(x) + C$  for some constant  $C$ .

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$A(x) = \int_a^x e^t dt$ . What functions could  $A(x)$  be?

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$A(x) = \int_a^x \cos t dt$ . What functions could  $A(x)$  be?

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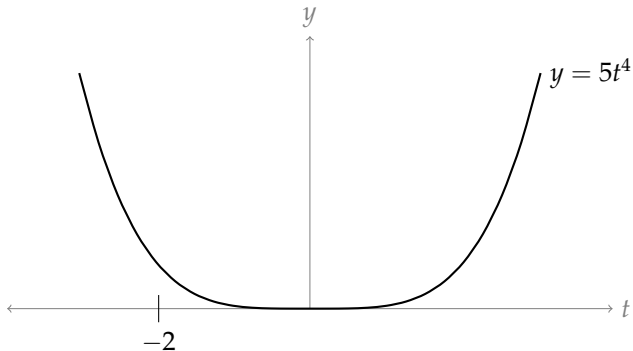
- ▶ If  $A(x) = \int_a^x f(t) dt$ , then<sup>1</sup>  $A'(x) = f(x)$
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$A(x) = \int_{-2}^x 5t^4 dt$ . What functions could  $A(x)$  be?

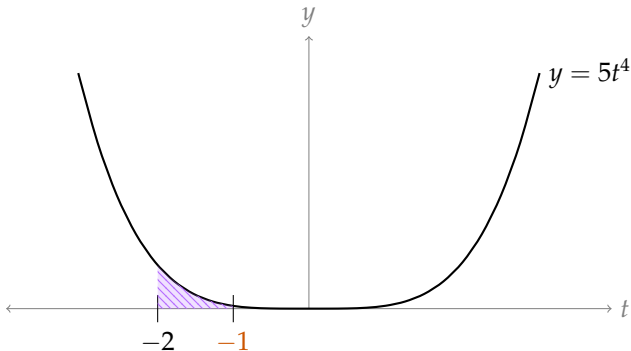
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<sup>1</sup>(as long as  $f(t)$  is continuous on  $[a, x]$ )

$$A(x) = \int_{-2}^x 5t^4 \, dt = x^5 + 32$$

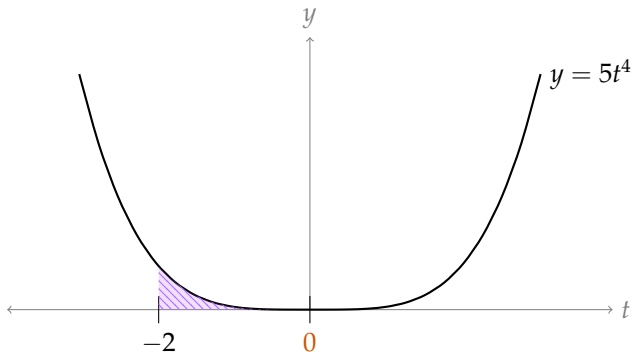


$$A(x) = \int_{-2}^x 5t^4 dt = x^5 + 32$$



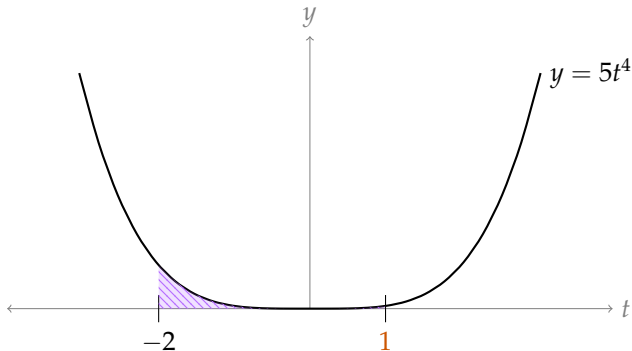
$$A(-1) = \int_{-2}^{-1} 5t^4 dt = (-1)^5 + 32 = 31$$

$$A(x) = \int_{-2}^x 5t^4 dt = x^5 + 32$$



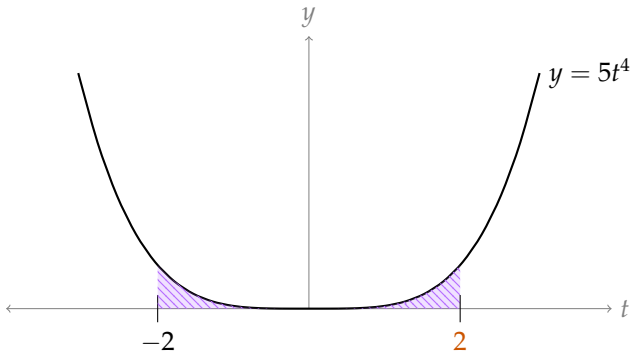
$$A(0) = \int_{-2}^0 5t^4 dt = (0)^5 + 32 = 32$$

$$A(x) = \int_{-2}^x 5t^4 \, dt = x^5 + 32$$



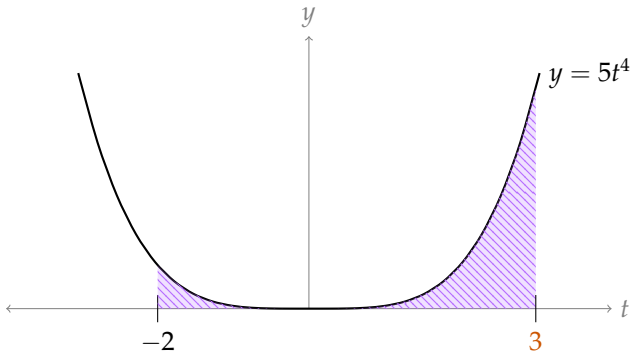
$$A(\textcolor{brown}{1}) = \int_{-2}^{\textcolor{brown}{1}} 5t^4 \, dt = (\textcolor{brown}{1})^5 + 32 = 33$$

$$A(x) = \int_{-2}^x 5t^4 dt = x^5 + 32$$



$$A(2) = \int_{-2}^2 5t^4 dt = (2)^5 + 32 = 64$$

$$A(x) = \int_{-2}^x 5t^4 dt = x^5 + 32$$



$$A(3) = \int_{-2}^3 5t^4 dt = (3)^5 + 32 = 275$$

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$A(x) = \int_a^x f(t) dt$ . What functions could  $A(x)$  be?

- ▶  $A'(x) = f(x)$ .
- ▶ Guess a function with derivative  $f(x)$ :  $F(x)$ .
- ▶ Then  $A(x) = F(x) + C$  for some constant  $C$ .
- ▶ Also  $A(a) = 0$ , so  $0 = F(a) + C$ , so  $C = -F(a)$
- ▶ So,  $A(x) = F(x) - F(a)$

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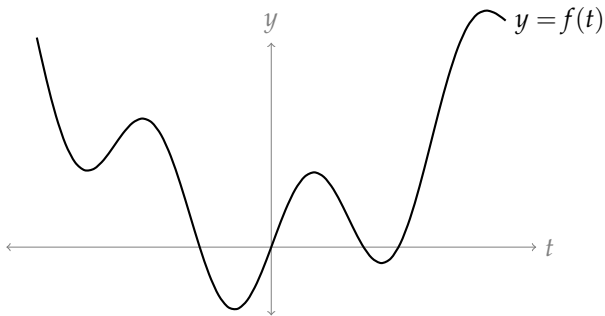
$A(b) = \int_a^b f(t) dt$ . What functions could  $A(b)$  be?

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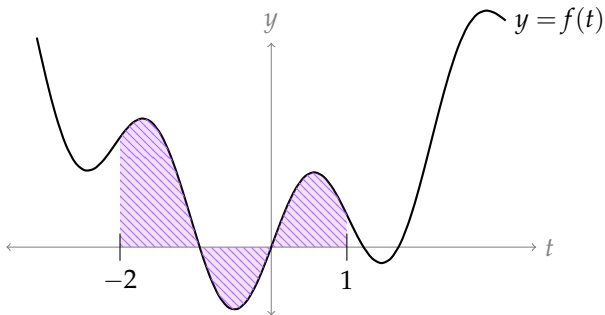
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$$\int_a^b f(t) \, dt = F(b) - F(a) \quad \text{where} \quad F'(x) = f(x)$$

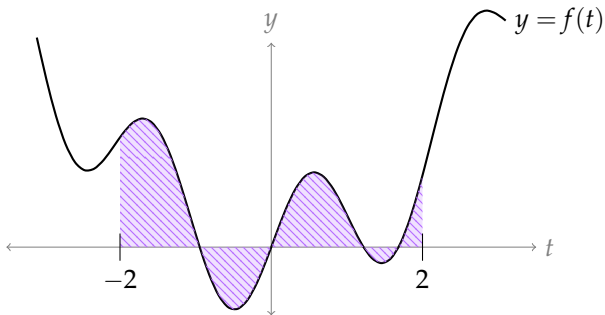


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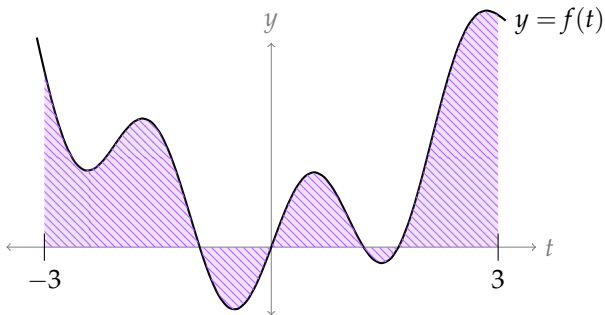
$$\int_{-2}^1 f(t) \, dt = F(1) - F(-2)$$

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$$\int_a^b f(t) \, dt = F(b) - F(a) \quad \text{where} \quad F'(x) = f(x)$$



$$\int_{-3}^3 f(t) \, dt = F(3) - F(-3)$$

## Fundamental Theorem of Calculus, Part 2

Let  $F(x)$  be differentiable, defined, and continuous on the interval  $[a, b]$  with  $F'(x) = f(x)$  for all  $a < x < b$ . Then

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Examples:

$$\frac{d}{dx} \{5x^7\} = 35x^6, \text{ so}$$

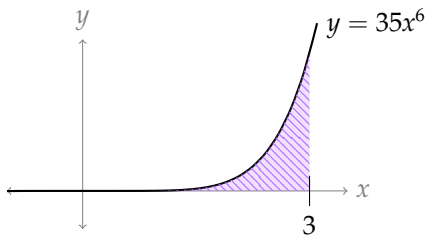
$$\int_0^3 35x^6 \, dx =$$

$$\frac{d}{dx} \{\tan x\} = \sec^2 x, \text{ so}$$

$$\int_0^{\pi/4} \sec^2 x \, dx =$$

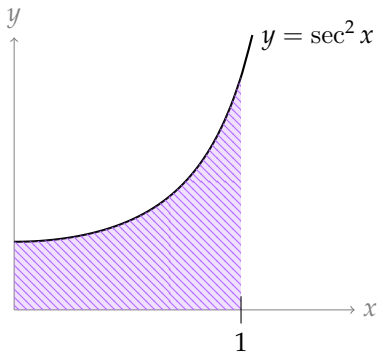


$$\int_0^3 35x^6 \, dx = F(b) - F(a) \quad \text{where} \quad F(x) = 5x^7$$



$$\int_0^3 35x^6 \, dx = 5(3)^7 - 5(0)^7$$

$$\int_0^{\pi/4} \sec^2 x \, dx = F(b) - F(a) \quad \text{where} \quad F(x) = \tan x$$



$$\int_0^{\pi/4} \sec^2 x \, dx = \tan\left(\frac{\pi}{4}\right) - \tan 0 = 1$$

# RELEVANT VOCABULARY

## Definition

If  $F(x)$  is a function whose derivative is  $f(x)$ , we call  $F(x)$  an **antiderivative** of  $f(x)$ .

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An antiderivative of  $\sin x$  is



# CONVENIENT NOTATION

## Definition

$$f(x) \Big|_a^b = f(b) - f(a)$$

The function  $f(x)$  evaluated from  $a$  to  $b$

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$$(5x + x^2) \Big|_1^2 =$$

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The function  $f(x)$  evaluated from  $a$  to  $b$

Examples:

$$(5x + x^2) \Big|_1^2 =$$

$$\frac{x^2}{x+2} \Big|_5^{-1} =$$

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## Definition

$$f(x) \Big|_a^b = f(b) - f(a)$$

The function  $f(x)$  evaluated from  $a$  to  $b$

## FTC Part 2, Abridged

$$\int_a^b f(x) \, dx = F(x) \Big|_a^b$$

where  $F(x)$  is an antiderivative of  $f(x)$

## Definition

The **indefinite integral** of a function  $f(x)$ :

$$\int f(x) \, dx$$

means the *most general* antiderivative of  $f(x)$ .

Examples:

$$\int 2x \, dx =$$

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Examples:

$$\int 2x \, dx =$$

$$\int \frac{1}{x} \, dx =$$

Remember: two functions with the same derivative differ by a constant, so we include the “+C” for indefinite integrals.

# DEFINITE VS INDEFINITE INTEGRALS

For each pair of properties below, decide which applies to **definite** integrals, and which to **indefinite** integrals.

|                                                       |  |
|-------------------------------------------------------|--|
| No limits (or bounds) of integration, $\int f(x) dx$  |  |
| Limits (or bounds) of integration, $\int_a^b f(x) dx$ |  |
| Area under a curve                                    |  |
| Antiderivative                                        |  |
| Number                                                |  |
| Function                                              |  |

# ANTIDIFFERENTIATION BY INSPECTION

1.  $\int e^x dx$

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1.  $\int e^x \, dx$

2.  $\int \cos x \, dx$

3.  $\int -\sin x \, dx$

4.  $\int \frac{1}{x} \, dx$

5.  $\int 1 \, dx$



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6.  $\int 2x \, dx$



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7.  $\int nx^{n-1} \, dx \quad (n \neq 0, \text{ constant})$



# ANTIDIFFERENTIATION BY INSPECTION

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5.  $\int 1 \, dx$

6.  $\int 2x \, dx$

7.  $\int nx^{n-1} \, dx \quad (n \neq 0, \text{ constant})$

8.  $\int x^n \, dx \quad (n \neq -1, \text{ constant})$



## Power Rule for Antidifferentiation

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C$$

if  $n \neq -1$  is a constant

Example:

$$\int (5x^2 - 15x + 3) dx =$$



# ANTIDERIVATIVES TO RECOGNIZE

- ▶  $\int x^n dx = \frac{1}{n+1}x^{n+1} + C$
- ▶  $\int a dx = ax + C$
- ▶  $\int e^x dx = e^x + C$
- ▶  $\int \frac{1}{x} dx = \log |x| + C$
- ▶  $\int \sin x dx = -\cos x + C$
- ▶  $\int \cos x dx = \sin x + C$
- ▶  $\int \sec^2 x dx = \tan x + C$
- ▶  $\int \sec x \tan x dx = \sec x + C$
- ▶  $\int \csc x \cot x dx = -\csc x + C$
- ▶  $\int \csc^2 x dx = -\cot x + C$
- ▶  $\int \frac{1}{1+x^2} dx = \arctan x + C$
- ▶  $\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$

## Included Work



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